

CareerAI: AI-Powered Career Guidance & Student Success Platform

Assignment 14: AI-Powered Career Guidance & Student Success Platform

Domain:

Education & Career Development

Prepared By:

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Technology Used:

n8n Automation, AI Models, Google Forms, Google Sheets, Gmail, Database

Project Type:

AI-Powered Automation Platform

Introduction

CareerAI is an AI-powered Career Guidance and Student Success Platform designed to help students make informed career decisions by providing personalized recommendations and continuous career development support.

In today's rapidly changing technology landscape, students need proper guidance to understand different career opportunities, industry requirements, and the skills required to achieve their goals. However, many students struggle to identify suitable career paths because they do not have access to personalized mentorship, updated industry information, or structured learning plans.

CareerAI solves this problem by using Artificial Intelligence and automation workflows to analyze student profiles, including academic background, technical skills, interests, certifications, and career aspirations. Based on this analysis, the platform provides customized career recommendations, identifies skill gaps, suggests relevant learning resources, and generates personalized career roadmaps.

The platform is built using **n8n automation workflows** that connect different services such as AI models, Google Forms, Google Sheets, Gmail, and databases. These automated workflows reduce manual effort and allow career guidance services to support a large number of students efficiently.

The system also helps career counselors by generating student progress reports, providing analytics, and offering data-driven insights for better decision-making.

Problem Analysis

Selecting the right career path is one of the most important decisions for students. However, many students face difficulties because they lack proper knowledge about available career options, industry trends, required technical skills, and future opportunities.

Students from different educational backgrounds often have different interests and abilities, but traditional career guidance systems usually provide generalized advice rather than personalized recommendations. This can result in students choosing unsuitable courses, developing irrelevant skills, or missing valuable internship and job opportunities.

Another major challenge is the limited availability of career counselors. A counselor may have to guide hundreds of students, making it difficult to analyze each student's:

- Academic performance
- Skills and technical knowledge
- Career interests
- Learning progress
- Professional goals

Manual career assessment requires significant time and effort, and it becomes difficult to maintain updated information about changing industry requirements.

The major problems identified are:

1. Lack of Personalized Career Guidance

Students often receive general advice that does not consider their individual skills, interests, and goals.

2. Skill Gap Between Education and Industry

Many students are unaware of the skills required for their desired careers, creating a gap between academic learning and industry expectations.

3. Difficulty Finding Learning Resources

Students struggle to identify the right courses, certifications, and projects needed for career growth.

4. Limited Internship and Job Awareness

Students may miss suitable opportunities because they do not know where and how to find relevant internships and jobs.

5. Manual Workload for Career Counselors

Counselors spend considerable time collecting student information, analyzing profiles, and preparing reports manually.

6. Lack of Progress Monitoring

Students and counselors often do not have a centralized system to track skill development and career readiness.

To overcome these challenges, CareerAI provides an automated AI-driven solution that supports students throughout their career planning journey.

Business Context

The education sector is experiencing a growing need for intelligent and scalable career guidance solutions. Universities and educational institutions are responsible for helping students

achieve successful career outcomes, but providing personalized guidance to every student is challenging due to limited resources and increasing student numbers.

Traditional counseling approaches depend heavily on manual interaction between students and career counselors. Although counselors provide valuable guidance, analyzing individual student profiles, tracking progress, and keeping recommendations updated requires significant time and effort.

CareerAI provides an automated solution that supports educational institutions by combining Artificial Intelligence with workflow automation. The platform enables institutions to provide personalized career support at scale while reducing the workload on counselors.

The system analyzes student data and generates meaningful insights related to:

- Suitable career paths
- Required technical and soft skills
- Skill gaps
- Recommended courses and certifications
- Internship and job opportunities
- Student progress and career readiness

By implementing AI-based decision-making and automated workflows, institutions can improve student engagement, enhance placement preparation, and provide better career development support.

Business Benefits

1. Scalable Career Support

The platform can provide personalized guidance to a large number of students without increasing manual workload.

2. Improved Student Success Rate

Students receive accurate career recommendations and structured learning plans based on their individual profiles.

3. Data-Driven Counseling

Counselors can use AI-generated reports and analytics to make better decisions.

4. Efficient Resource Management

Automation reduces repetitive tasks such as data collection, analysis, and report generation.

5. Better Industry Alignment

Students can develop skills according to current market requirements and career opportunities.

Stakeholders Analysis

The CareerAI platform involves multiple stakeholders who benefit from the system.

Stakeholder	Role	Benefits
Students	Primary users of the platform	Receive personalized career recommendations, skill analysis, learning roadmaps, and internship suggestions

Stakeholder	Role	Benefits
Career Counselors	Monitor and guide students	Get AI-generated reports, student analytics, and career insights
Educational Institutions	Implement and manage the platform	Improve student outcomes, placement readiness, and counseling efficiency
Faculty Members	Support student development	Track student progress and identify areas requiring improvement
Recruiters/Companies	Provide career opportunities	Connect with students having relevant skills and career preparation
Administrators	Manage platform operations	Monitor system performance and generate institutional reports

Pain Points

Student Pain Points

Lack of Career Awareness

Students often do not have complete information about different career options and future opportunities.

Difficulty Identifying Required Skills

Students may know their desired career but are unaware of the technical and professional skills required.

Confusion About Learning Resources

Selecting appropriate courses, certifications, and projects can be difficult due to the availability of many online resources.

Limited Access to Mentorship

Many students do not have continuous access to professional career guidance.

Missing Opportunities

Students may miss internships, competitions, and job opportunities due to lack of awareness.

Career Counselor Pain Points

High Number of Students

Counselors often handle many students, making personalized guidance difficult.

Manual Data Analysis

Collecting and analyzing student information requires considerable time.

Difficulty Tracking Progress

Monitoring each student's learning journey and improvement is challenging.

Limited Real-Time Insights

Counselors may not have updated information about student skills and career readiness.

Institution Pain Points

Lack of Automation

Many career guidance activities are performed manually.

Difficulty Measuring Career Development

Institutions need better methods to evaluate student progress and placement readiness.

Resource Limitations

Providing individual career counseling to every student requires significant resources.

Objectives

The main objective of CareerAI is to develop an intelligent and automated career guidance platform that helps students make better career decisions through AI-based analysis and personalized recommendations.

The key objectives are:

1. Provide Personalized Career Recommendations

The platform analyzes student information such as academic background, skills, interests, certifications, and career goals to recommend suitable career paths.

Example:

A student interested in:

- Python

- Artificial Intelligence
- Data Analysis

may receive recommendations such as:

- Generative AI Engineer
- Machine Learning Engineer
- Data Scientist

2. Identify Skill Gaps

The system compares the student's current skills with the skills required for their desired career path.

It identifies:

- Missing technical skills
- Required certifications
- Recommended learning areas
- Improvement opportunities

This helps students understand what they need to learn to achieve their career goals.

3. Generate Personalized Learning Roadmaps

CareerAI creates structured learning plans based on student goals.

The roadmap includes:

- Recommended courses

- Learning resources
- Certifications
- Projects
- Skill development timeline

This provides students with a clear path for career preparation.

4. Recommend Internships and Job Opportunities

The platform helps students discover relevant opportunities based on their:

- Skills
- Career interests
- Education background
- Experience level

It improves awareness of internships, training programs, and job opportunities.

5. Track Student Progress

The system monitors student development by analyzing:

- Completed courses
- Developed skills
- Certifications earned
- Learning progress

It generates progress reports to measure career readiness.

6. Support Career Counselors

The platform provides AI-generated reports and analytics that help counselors understand student requirements and provide better guidance.

Proposed Solution

CareerAI is an AI-powered automated career guidance platform that uses n8n workflows to connect different services and automate the complete career planning process.

The platform collects student information through digital forms and processes this data using Artificial Intelligence models. The AI system analyzes student profiles and generates personalized career recommendations, skill gap analysis, learning roadmaps, and opportunity suggestions.

The solution consists of multiple independent automation workflows:

Workflow 1: Student Registration & Career Assessment

Collects student information including:

- Personal details
- Academic performance
- Skills
- Interests
- Career goals

The collected data is stored securely for further processing.

Workflow 2: AI Career Recommendation Engine

Uses AI models to analyze student profiles and recommend suitable career paths based on their abilities and interests.

Workflow 3: Learning Roadmap Generation

Creates personalized learning plans by identifying required skills and recommending courses, certifications, and projects.

Workflow 4: Internship & Job Recommendation

Analyzes student profiles and recommends suitable internships and job opportunities.

Workflow 5: Counselor Review & Progress Report

Generates automated reports that help counselors monitor student progress and provide effective guidance.

8. Key Features

AI-Based Career Analysis

The platform uses Artificial Intelligence to understand student profiles and provide intelligent career recommendations.

Skill Gap Detection

Identifies missing skills by comparing current student abilities with industry requirements.

Personalized Learning Roadmap

Generates customized learning paths containing:

- Courses
- Certifications
- Projects
- Skill development plans

Internship and Job Recommendations

Provides relevant career opportunities based on student profiles.

Automated Progress Reports

Generates reports for students and counselors to track career development.

Workflow Automation

Uses n8n to automate:

- Data collection
- AI processing
- Notifications
- Reports
- Decision-making processes

9. Technology Stack

CareerAI uses a combination of Artificial Intelligence, workflow automation, cloud-based tools, and data management technologies to build an efficient career guidance platform.

n8n Automation Platform

n8n is used as the core automation engine of the system. It connects different services and manages the complete workflow process, including data collection, AI processing, notifications, and report generation.

Artificial Intelligence Models (Gemini)

AI models are used for intelligent decision-making and analysis. They analyze student profiles, recommend suitable career paths, identify skill gaps, and generate personalized learning roadmaps.

Google Forms

Google Forms is used to collect student information such as academic details, skills, interests, certifications, and career goals.

Google Sheets

Google Sheets is used for storing and managing student data, career recommendations, workflow outputs, and progress information.

Gmail Integration

Gmail automation is used to send notifications, career recommendations, learning roadmaps, internship updates, and progress reports to students and counselors.

Database

A database is used to maintain structured records of student profiles, AI-generated recommendations, learning progress, and career reports.

Napkin

These tools are used to create system architecture diagrams, workflow interaction diagrams, and event flow representations.

GitHub

GitHub is used for project documentation, version control, storing workflow JSON files, and managing project resources.

System Architecture & Workflow Diagrams

1. Overall System Architecture

System Architecture Description

CareerAI follows a modular architecture where student data is collected, processed through AI models, stored in databases, and delivered as personalized career guidance.

The system consists of five major layers:

1. User Layer

This layer includes students and career counselors who interact with the platform.

Students provide:

- Academic details
- Skills
- Interests

- Career goals
- Certifications

Counselors receive:

- Student progress reports
- Career analytics
- AI-generated insights

2. Data Collection Layer

This layer collects student information through:

- Google Forms
- **Webhooks**

The collected data is transferred to n8n workflows for processing.

3. Automation Layer (n8n)

n8n acts as the central automation engine that manages all workflows.

It performs:

- Data processing
- Workflow execution
- AI integration
- Conditional decision-making
- Notifications
- Report generation

The five workflows connected with n8n are:

1. Student Registration & Career Assessment

2. AI Career Recommendation Engine
3. Learning Roadmap Generation
4. Internship & Job Recommendation
5. Counselor Progress Report

4. AI Processing Layer

The AI layer performs intelligent analysis using AI models.

Functions:

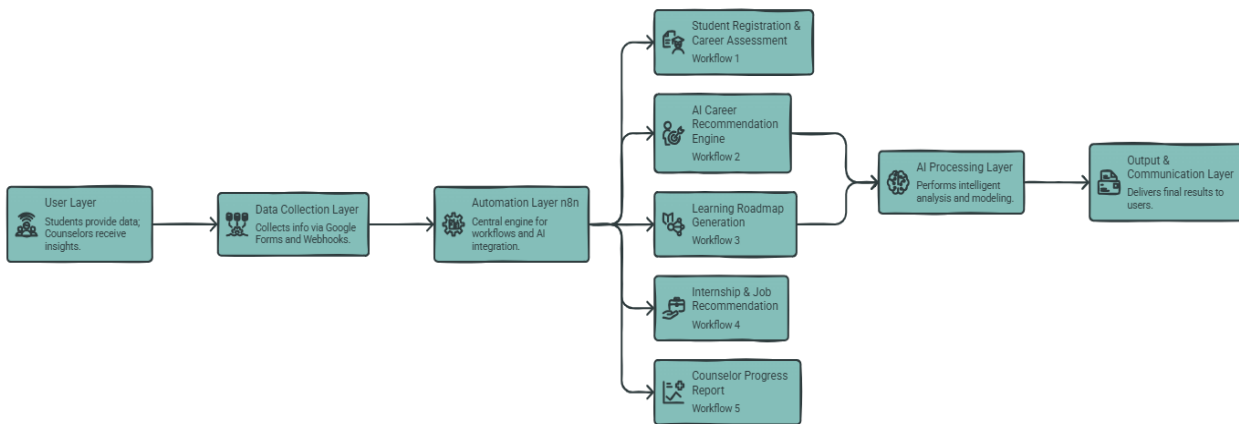
- Career recommendation
- Skill gap identification
- Learning roadmap generation
- Opportunity filtering
- Progress analysis

5. Output & Communication Layer

The final results are delivered through:

- Email notifications
- Reports
- Career recommendations
- Learning plans
- Internship suggestions

CareerAI System Architecture



Made with Napkin

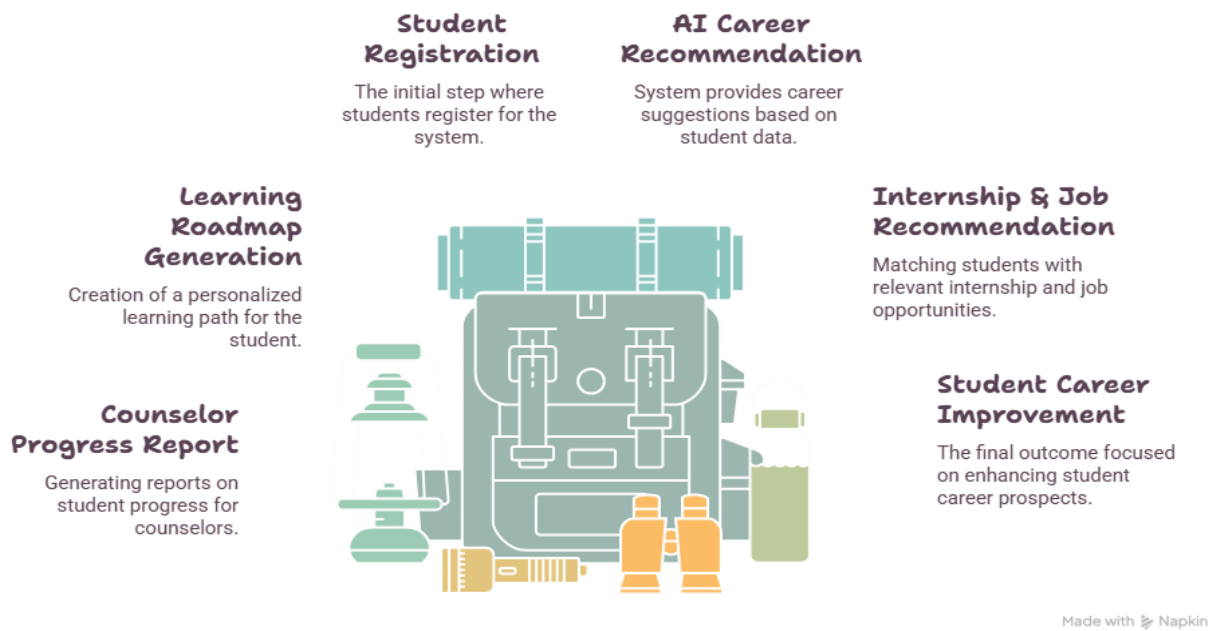
2. Workflow Interaction Diagram

This diagram shows how your 5 workflows communicate.

Description

The CareerAI platform follows a sequential workflow approach. Student information is first collected, then analyzed by AI, followed by roadmap generation, opportunity recommendations, and counselor reporting.

Career Development Workflow



3. Event Flow Between Workflows

Description

Each workflow is triggered by specific events and passes information to the next workflow.

Event Flow:

Career Assessment and Planning Workflow



n8n Workflow Documentation

Workflow 1: Student Registration & Career Assessment

Purpose

The Student Registration & Career Assessment workflow is the entry point of the CareerAI platform. It collects student information through an online form and creates a digital student profile that is used throughout the career guidance process.

This workflow ensures that all necessary academic and personal information is captured accurately before AI-based analysis begins.

Trigger

The workflow is triggered when a student submits the career assessment form using Google Forms or a Webhook.

Input

The workflow collects the following information:

- Student Name
- Email Address
- College/University
- Degree and Branch
- Academic Year
- CGPA or Percentage

- Technical Skills
- Areas of Interest
- Certifications
- Career Goal

Workflow Process

The workflow performs the following steps:

1. Receives student details from the registration form.
2. Validates the submitted information.
3. Stores the student profile in Google Sheets or a database.
4. Creates a unique student record.
5. Sends a confirmation email acknowledging successful registration.
6. Makes the student profile available for AI analysis in the next workflow.

Output

After successful execution, the workflow generates:

- Registered student profile
- Stored student data
- Confirmation email
- Input data for the AI Career Recommendation workflow

Tools Used

- Google Forms
- Gemini
- n8n
- Google Sheets / Database
- Gmail

Expected Outcome

The workflow successfully creates a complete student profile that becomes the foundation for all future recommendations and career planning activities.

Workflow 2: AI Career Recommendation Engine

Purpose

This workflow analyzes the student's academic background, skills, interests, certifications, and career goals using Artificial Intelligence. Based on this analysis, it recommends suitable career paths and provides personalized career guidance.

Trigger

The workflow starts automatically after the student registration workflow successfully stores the student profile.

Input

The workflow receives:

- Student profile
- Academic details
- Technical skills
- Interests
- Certifications
- Career goals

Workflow Process

The workflow performs the following operations:

1. Retrieves the student's profile from the database or Google Sheets.
2. Sends the information to the AI model (Gemini).
3. Analyzes the student's strengths and interests.
4. Identifies suitable career options.
5. Determines the required skills for each recommended career.
6. Stores the AI-generated recommendations.

Output

The workflow generates:

- Personalized career recommendations
- Career suitability analysis

- Required skills
- Career readiness insights

Tools Used

- n8n
- OpenAI/Gemini API
- Google Sheets / Database

Expected Outcome

Students receive AI-generated career recommendations that match their skills, interests, and career aspirations.

Workflow 3: Learning Roadmap Generation

Purpose

The Learning Roadmap Generation workflow creates a personalized learning plan for each student based on the career recommendations provided by the AI. It identifies the skills required for the student's desired career and suggests a structured roadmap to help them achieve their goals.

The roadmap enables students to learn systematically by recommending relevant courses, certifications, projects, and skill development activities.

Trigger

This workflow is triggered automatically after the AI Career Recommendation workflow generates career recommendations for the student.

Input

The workflow receives:

- Student profile
- Career recommendations
- Current skills
- Career goals
- AI analysis results

Workflow Process

The workflow performs the following steps:

1. Receives the student's career recommendation.
2. Compares the student's current skills with industry requirements.
3. Identifies missing skills and knowledge areas.

4. Uses Google Gemini AI to generate a personalized learning roadmap.
5. Recommends online courses, certifications, and hands-on projects.
6. Stores the roadmap and sends it to the student.

Output

The workflow generates:

- Personalized learning roadmap
- Recommended courses
- Certification suggestions
- Project recommendations
- Skill development plan

Tools Used

- n8n
- Google Gemini API
- Google Sheets / Database
- Gmail

Expected Outcome

Students receive a customized learning roadmap that helps them develop the necessary skills and prepare for their desired career.

Workflow 4: Internship & Job Recommendation

Purpose

The Internship & Job Recommendation workflow helps students discover relevant internship and job opportunities based on their career interests, technical skills, and AI-generated career recommendations.

The workflow ensures that students receive opportunities aligned with their qualifications and career aspirations.

Trigger

This workflow is triggered after the Learning Roadmap Generation workflow is completed.

Input

The workflow receives:

- Student profile
- Career recommendations
- Skills
- Learning roadmap
- Career goals

Workflow Process

The workflow performs the following steps:

1. Retrieves the student's career profile.
2. Analyzes the student's skills and interests.
3. Uses Google Gemini AI to identify suitable opportunities.
4. Filters internship and job recommendations based on the student's profile.
5. Stores the recommendations.
6. Sends the recommendations to the student through email.

Output

The workflow generates:

- Internship recommendations
- Job recommendations
- Application guidance
- Career opportunity report

Tools Used

- n8n
- Google Gemini API
- Google Sheets / Database
- Gmail

Expected Outcome

Students receive internship and job recommendations that match their career goals and current skill set.

Workflow 5: Counselor Review & Progress Report

Purpose

The Counselor Review & Progress Report workflow enables career counselors to monitor student progress through AI-generated reports and analytics. It helps counselors identify strengths, skill gaps, and learning progress, allowing them to provide personalized guidance.

Trigger

This workflow is triggered after the Internship & Job Recommendation workflow or on a scheduled basis using a Cron trigger.

Input

The workflow receives:

- Student profile
- Career recommendations
- Learning roadmap
- Internship and job recommendations
- Progress data

Workflow Process

The workflow performs the following steps:

1. Collects the student's latest progress data.
2. Analyzes learning achievements and completed activities.
3. Uses Google Gemini AI to generate a comprehensive progress report.
4. Creates performance insights and career readiness analysis.
5. Stores the report in the database.
6. Sends the report to the career counselor and, if required, to the student.

Output

The workflow generates:

- Student progress report
- Career readiness analysis
- Skill improvement summary
- Counselor recommendations

Tools Used

- n8n
- Google Gemini API
- Google Sheets / Database

- Gmail
- Cron (if scheduled)

Advanced Features & Implementation

AI-Powered Decision Making (Implemented)

The CareerAI platform uses the **Google Gemini API** to analyze student profiles and generate personalized career guidance. The AI evaluates academic performance, technical skills, interests, certifications, and career goals to recommend suitable career paths.

It also identifies skill gaps and generates personalized learning roadmaps, helping students make informed career decisions based on their individual profiles.

Benefits

- Personalized career recommendations
- AI-based skill gap analysis
- Customized learning roadmap
- Improved career planning

Conditional Branching (Implemented)

The platform uses conditional logic within the n8n workflows to process student information based on different inputs.

For example, recommendations vary depending on:

- Student interests
- Technical skills
- Career goals
- Academic background

This ensures that each student receives recommendations relevant to their profile.

Benefits

- Dynamic workflow execution
- Personalized recommendations
- Efficient automation

Future Enhancements

The following features can be added in future versions of the platform to further improve automation and reliability:

Human Approval

A counselor approval step can be introduced before AI-generated recommendations are shared with students. This would allow counselors to review and modify recommendations when necessary.

Error Handling and Retry Logic

Future versions can include automatic error handling and retry mechanisms to manage API failures, email delivery issues, or database connection problems.

Logging and Audit Trail

Workflow execution logs can be maintained to record:

- Workflow status
- Execution time
- Error messages
- Student activity

This would improve monitoring and troubleshooting.

Project Implementation Summary

The CareerAI platform was successfully developed using **n8n workflow automation** and the **Google Gemini API**. The project consists of five independent workflows that automate the complete career guidance process, from student registration to counselor reporting.

The implemented system provides AI-powered career recommendations, learning roadmap generation, internship guidance, and progress reporting. Conditional workflow logic ensures that recommendations are personalized based on each student's profile.

The modular architecture makes the platform scalable and allows additional features such as human approval, scheduled workflows, webhook integration, and advanced error handling to be incorporated in future versions.

Testing & Results

Testing Methodology

To verify the functionality of the CareerAI platform, each workflow was tested independently as well as in sequence. Sample student data was used to evaluate the accuracy of AI-generated career recommendations, learning roadmaps, internship suggestions, and counselor reports.

Testing ensured that the workflows executed correctly and that data flowed successfully between the different components of the system.

The testing process focused on:

- Student registration and data collection
- AI-based career recommendation
- Learning roadmap generation
- Internship and job recommendation
- Progress report generation

Test Case 1: Student Registration

Objective: Verify that student information is collected and stored successfully.

Input:

- Student Name: Simran Kumari
- Degree: B.Tech Computer Science
- Year: Second Year
- Skills: Python, HTML, CSS, JavaScript
- Interest: Artificial Intelligence
- Career Goal: Generative AI Engineer

Expected Result:

The student's information should be successfully stored and made available for AI analysis.

Actual Result:

The workflow executed successfully, and the student profile was stored for further processing.

Status: Passed

Test Case 2: AI Career Recommendation

Objective: Verify AI-generated career recommendations.

Input:

Student profile generated from Workflow 1.

Expected Result:

The AI should recommend suitable career options based on the student's skills and interests.

Actual Result:

Google Gemini successfully analyzed the profile and generated personalized career recommendations along with required skills.

Status: Passed

Test Case 3: Learning Roadmap Generation

Objective: Verify that the system generates a personalized learning roadmap.

Expected Result:

The platform should recommend courses, certifications, and projects based on the selected career path.

Actual Result:

A customized learning roadmap was successfully generated for the student.

Status: Passed

Test Case 4: Internship & Job Recommendation

Objective: Verify internship and job recommendations.

Expected Result:

The workflow should recommend relevant internship and job opportunities according to the student's profile.

Actual Result:

The system successfully generated recommendations aligned with the student's skills and career goals.

Status: Passed

Test Case 5: Counselor Progress Report

Objective: Verify report generation.

Expected Result:

The workflow should generate a progress report summarizing the student's career development.

Actual Result:

The counselor report was successfully generated and included career recommendations, learning progress, and improvement suggestions.

Status: Passed

Results

The CareerAI platform successfully automated the career guidance process using five independent n8n workflows and Google Gemini AI. The system collected student information, generated AI-powered career recommendations, created personalized learning roadmaps, suggested internship opportunities, and produced counselor reports.

The implementation demonstrated that workflow automation and Artificial Intelligence can significantly improve the efficiency of career guidance while providing personalized support to students.

Key Achievements

- Successfully automated the career guidance process.

- Generated AI-powered personalized career recommendations.
- Created customized learning roadmaps based on student profiles.
- Recommended relevant internship and career opportunities.
- Generated counselor progress reports for monitoring student development.
- Reduced manual effort through workflow automation.

Challenges Faced

During the development of the CareerAI platform, several challenges were encountered:

- Integrating multiple services within n8n.
- Designing workflows that effectively pass data between stages.
- Writing prompts to obtain accurate responses from Google Gemini.
- Ensuring proper data flow and conditional logic.
- Testing workflows with different student profiles to verify consistent results.

These challenges were addressed through workflow testing, prompt refinement, and iterative improvements.

Future Scope

The CareerAI platform can be enhanced with additional features to provide more comprehensive career guidance services.

Future improvements include:

- Integration with LinkedIn and Internshala for live internship and job listings.
- Real-time skill assessment and aptitude testing.
- Human approval workflow for counselor verification.
- Cron-based scheduled reports and reminder notifications.
- Webhook integration for real-time external applications.
- Dashboard for students and counselors with analytics and progress tracking.
- Mobile application support for easier accessibility.
- Advanced analytics to monitor placement readiness and institutional performance.

Conclusion

CareerAI demonstrates how Artificial Intelligence and workflow automation can transform traditional career guidance into a personalized, scalable, and efficient system. By integrating Google Gemini AI with n8n workflows, the platform provides intelligent career recommendations, learning roadmaps, internship suggestions, and progress reports based on individual student profiles.

The modular architecture makes the platform easy to maintain and extend, while its AI-driven approach helps students make informed career decisions and supports counselors with

meaningful insights. Overall, the project showcases the practical application of AI and automation in the field of education and career development.